

Pre-conditions Report Template

CS 789 - Software Reengineering

Milestone 1 | Due Tuesday, October 13, 2026

Team: _____

Repository: _____

Team member name	UNLV email address

List every team member. Leave the fourth row blank for a three-person team.

Purpose: demonstrate that the team is ready to begin the project and that every student has independently prepared to work on their own assigned PRs.

Read the complete project instructions before using this template. The project instructions define the required activities; this template structures the evidence to report at this milestone.

The three reports build on one another. The Pre-conditions Report establishes readiness and the initial plan. The Intermediate Report reports actual reengineering progress. The Final Report consolidates the complete project.

Milestone	Main question	Progression
Pre-conditions	Are you ready to begin?	Build -> understand -> allocate -> inspect -> plan
Intermediate	Are you carrying out the process?	Analyze -> recover design -> integrate -> test -> adjust
Final	What happened across the project?	Complete -> validate -> compare -> synthesize -> reflect
Individual responsibility: Although the team submits one report, each student is individually responsible for preparing, analyzing, integrating, testing, and documenting their assigned PRs. Sections marked [INDIVIDUAL] must be completed by the named student.		

1. Team and Project Setup [TEAM]

- Confirm that the team information and exact GitHub repository/fork URL above are complete and correct.
- Confirm that the target is linkedin/kafka on branch 3.0-li and that the source PRs come from apache/kafka.
- Briefly state the reengineering problem: integrating changes from an upstream system into a long-lived, diverged variant.
- Describe how the team will communicate, share findings, and keep individual PR work identifiable.

2. Reengineering Strategy and Project Plan [TEAM]

2.1 Scope and Goals

State what the team intends to accomplish across the four PR categories.

2.2 Definition of Done

Define when an assigned PR is considered complete, including how a well-supported Category 4 failure will be treated.

2.3 Reengineering Workflow

Describe the common workflow: understand the change, recover source and target design, identify the gap, attempt integration, adapt where needed, test, and document the outcome.

2.4 Initial Patterns and Maxims

Identify a small set of OORP patterns/maxims that will guide the work and explain why they are relevant.

2.5 Milestones and Coordination

Your team must maintain regular and visible coordination throughout the project. The purpose is to ensure steady progress, identify problems early, and allow the instructor and TAs to follow the team's work between project milestones.

- Weekly team meetings. Hold at least one team meeting every week from the start of the project through submission of the Final Report. Meetings should focus on actual project progress, technical problems, decisions, and the work planned for the following week.
- Meeting minutes. Record brief minutes for every weekly meeting using the template at the end of this document. The minutes are not only for the team's internal use. They must provide enough information for the instructor and TAs to understand what was discussed, what each student has accomplished, what problems the team is facing, and what each student has committed to do next.
- Discord communication. Each team will have a dedicated Discord channel. Use it for ongoing coordination between meetings, including progress updates, questions, technical findings, and blockers. Important decisions made asynchronously should be reflected in the next meeting minutes so that there is a clear project record.
- Visibility and accountability. Keep the meeting minutes in one shared document accessible to the instructor and TAs throughout the project. Update the document after each meeting; do not wait until a report deadline to reconstruct the minutes. Individual progress must be recorded by name so that participation and outstanding work are visible throughout the project.
- Raise technical blockers early. Waiting until a submission deadline to begin substantial work or to seek help from teammates or the TAs is not an acceptable project strategy. Teams should aim to complete work early enough to leave time for troubleshooting, validation, and report consolidation.

Agree on a recurring meeting day/time where possible and state it below. Also state how the team will track PR progress toward the Intermediate and Final Reports.

Weekly meeting day/time: _____ Shared minutes link:

Team Discord channel: _____ Progress tracking method:

3. PR Allocation [TEAM]

List every assigned PR before individual analysis begins. Each student is responsible for seven PRs: 1 Category 1, 2 Category 2, 1 Category 3, and 3 Category 4.

Student	PR	Category	Expected integration path	Initial status
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned

[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned
[Name]	[#]	[1-4]	[Cherry-pick / Cherry-pick -> RePatch / diagnostic]	Assigned

Add rows until all PRs for all team members are listed.

4. Individual Readiness and PR Pre-analysis [INDIVIDUAL]

Repeat Section 4 for every student. Each student must establish their own working baseline and complete a preliminary investigation of all seven assigned PRs. The purpose is to show individual readiness before the deeper analysis required in the Intermediate Report.

4.1 Student: [Name]

A. Development Environment and Baseline Validation

Independently demonstrate that you can build and test the unmodified `linkedin/kafka:3.0-li` target variant on the environment you will use for the project. Follow the setup and build guidance in the project instructions and the repository documentation. Review Appendix A of the project for the Java environment and testing/coverage considerations.

- Record the target branch, baseline commit SHA, development environment, build result, tests executed, and test results.
- Document any build or test problems that already exist in the unmodified baseline.
- Include concise, readable evidence showing that the build and tests were actually executed in your environment and showing the outcome. The evidence must be attributable to you and to the baseline version used.
- Do not paste pages of terminal output. Include only the evidence needed to establish the baseline.

Item	Your evidence
Target branch	
Baseline commit SHA	
Development environment	
Build result	
Test(s) executed	
Test result	
Existing build/test issues, if any	
Evidence / figure reference	

B. Preliminary Analysis of Your Seven PRs

Inspect all seven assigned PRs at a preliminary level. You are not expected to complete integration or full design recovery at this milestone. The goal is to understand what each PR changes, where it will integrate, and which cases may require more effort.

PR	Cat.	Purpose / change	Affected files or components	Initial risk / observations	Planned approach

C. Initial Repository and Design Understanding

- Identify the merge commit and changed files for each PR, and note relevant repository history where it helps explain the change.
- Identify the principal source classes/components affected and locate the corresponding area in the target variant.
- Note obvious source-target differences that may affect integration. Full design recovery is not required yet.

D. Initial Risk, Effort, and Prioritization

- Identify which PRs appear straightforward, difficult, or exceptional and briefly explain why.
- State which PRs you plan to investigate first and provide a rough effort estimate for your assigned work.

E. Planned Techniques and Patterns

- Identify the lab techniques/tools and OORP patterns that appear most useful for your assigned PRs and briefly justify the choices.

5. Consolidated Team Plan [TEAM]

After every student completes the individual readiness and pre-analysis, consolidate the findings into a short team plan.

- Identify the PRs or components that appear highest risk and any recurring issues across students.
- Explain where team members may share knowledge or troubleshooting while retaining individual ownership of assigned PRs.
- Update the planned schedule if the individual pre-analysis revealed unexpected difficulty.

5.1 Weekly Meeting Minutes Template

Maintain the minutes in one shared document that remains accessible to the instructor and TAs throughout the project. Add one record after every weekly meeting so that project progress can be followed as it happens, rather than reconstructed at a milestone deadline.

Meeting date / time	
Attendees / absences	Identify members present and absent
Progress since last meeting	What each member completed, by name
Current work	What each member is working on now
Problems / blockers	Technical or coordination problems and help needed
Decisions / shared findings	Important decisions, discoveries, or approaches discussed
Next actions	Owner + target date for each action
Next meeting	Planned date / time

Keep the minutes concise, but specific enough to show what was discussed, each member's progress, current blockers, decisions, and individual commitments for the next week. Progress should be attributed to students by name.

6. What Comes Next

Intermediate Report - November 4, 2026

Every student must move beyond planning and provide substantive reengineering work on at least four of their seven assigned PRs, including at least one Category 4 PR. All seven PRs must remain visible in the project status, with preliminary investigation and a concrete next step for PRs not yet analyzed substantively.

The report should show actual repository evidence, source and target design recovery, integration/RePatch results where applicable, testing evidence, meaningful use of lab techniques, and current blockers or remaining work.

Final Report - December 2, 2026

The Final Report is the consolidated account of the entire project. Every student must complete and document all seven assigned PRs and complete both LLM-assisted activities described in the project instructions. The team then synthesizes findings across PRs and LLM experiments, evaluates tools and OORP patterns, and reflects on the complete project.